

Name and Family Name: Hamid Reza Ghahramani

Part1

Background

Relapse (عود) refers to the return of drug-seeking behavior or substance use after a period of abstinence or recovery. Relapse is a complex phenomenon influenced by changes in brain circuits involved in reward, motivation, learning, stress response, and decision-making. Recent studies have shown that epigenetic mechanisms, such as DNA methylation, histone modification, and regulation of gene expression, may contribute to long-lasting changes in these brain regions and increase vulnerability to relapse.

What is the role of epigenetic mechanisms in brain regions involved in relapse vulnerability among individuals with substance use disorders?

1. Define the PICO / PECO Elements

Break down your research question into its core elements:

- **P (Population/Patient/Problem)** individuals with substance use disorders
- **I / E (Intervention / Exposure)** epigenetic mechanisms
- **C (Comparison, if any)** Normal activity of brain regions
- **O (Outcome related to brain imaging)** in brain regions Activity involved in relapse vulnerability

2. Identify the Main Concepts

Extract the key concepts from your PICO/PECO elements that will be used in your literature search.

3. Create a Concept Table

Use PubMed (MeSH) and Embase (EMTREE) to find controlled vocabulary terms, and also collect synonyms, alternate spellings, and related terms.

Concepts	MeSH Term	Emtree	Subjects	Synonyms
Substance Use Disorders	(((((Drug Use Disorders[MeSH Terms]) OR (Substance Use Disorders[MeSH Terms]))) OR (Organic	'addict'/exp OR 'addict' OR 'addiction, drug'/exp OR 'addiction, drug' OR 'dependence,	DE "Substance Use Disorder Medications" OR DE "Substance Use	"addict" OR "addiction, drug" OR "dependence, drug" OR "drug addict" OR "drug addiction" OR "drug dependence

	Mental Disorders, Substance-Induced[MeSH Terms])) OR (Chemical Dependence[MeSH Terms])) OR (Disorder, Substance Related[MeSH Terms])) OR (Abuse, Drug[MeSH Terms])) OR (Prescription Drug Abuse[MeSH Terms])) OR (Drug Addiction[MeSH Terms]))	drug'/exp OR 'dependence, drug' OR 'drug addict'/exp OR 'drug addict' OR 'drug addiction'/exp OR 'drug addiction' OR 'drug dependence model'/exp OR 'drug dependence model' OR 'drug dependency'/exp OR 'drug dependency' OR 'drug facilitation'/exp OR 'drug facilitation' OR 'drug habituation'/exp OR 'drug habituation' OR 'drug physical dependence'/exp OR 'drug physical dependence' OR 'drug usage disorder'/exp OR 'drug usage disorder' OR 'drug use disorder'/exp OR 'drug use disorder' OR 'physical dependence'/exp OR 'physical dependence' OR 'substance addiction'/exp OR 'substance addiction' OR 'substance dependence'/exp OR 'substance dependence' OR 'substance dependency'/exp OR 'substance dependency' OR 'substance usage disorder'/exp OR 'substance usage disorder' OR 'substance use	Disorder" OR DE "Substance Use Prevention" OR DE "Addiction Counselors" OR DE "Drug Abuse" OR DE "Substance Use Treatment" OR DE "Substance Use and Addiction Measures"	model" OR "drug dependency" OR "drug facilitation" OR "drug habituation" OR "drug physical dependence" OR "drug usage disorder" OR "drug use disorder" OR "physical dependence" OR "substance addiction" OR "substance dependence" OR "substance dependency" OR "substance usage disorder" OR "substance use disorder" OR "substance use disorders" OR "substance-related disorder" OR "substance-related disorders" OR "toxicomania" OR "toxicomanias" OR "toxicomanie" OR "drug dependence"
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		disorder'/exp OR 'substance use disorder' OR 'substance use disorders'/exp OR 'substance use disorders' OR 'substance-related disorder'/exp OR 'substance-related disorder' OR 'substance-related disorders'/exp OR 'substance-related disorders' OR 'toxicomania'/exp OR 'toxicomania' OR 'toxicomanias'/exp OR 'toxicomanias' OR 'toxicomanie'/exp OR 'toxicomanie' OR 'drug dependence'/exp OR 'drug dependence'		
Epigenetic Mechanisms	((Epigenomic[MeSH Terms]) OR (Epigenetics[MeSH Terms])) OR (Epigenetic[MeSH Terms])	'epigenetics'/exp	DE "Epigenetics" OR DE "Biosynthesis" OR DE "Precision Medicine" OR DE "Selective Breeding" OR DE "Genetic Dominance"	"Epigenetics"
Normal activity of brain regions	(Encephalon[MeSH Terms]) OR (Brain[MeSH Terms])	'cerebra (brain)'/exp OR 'cerebra (brain)' OR 'cerebrum'/exp OR 'cerebrum' OR 'encephalon'/exp OR 'encephalon' OR 'brain'/exp OR 'brain'	DE "Brain" OR DE "Brain Concussion" OR DE "Brain Lesions (Experimental)" OR DE "Brain Computer Interface" OR DE "Brain Health"	"cerebra (brain)" OR "cerebrum" OR "encephalon" OR "brain"
Relapse Vulnerability	(((((Recurrences[MeSH Terms]) OR (Relapse[MeSH	'relapse'/exp	DE "Relapse Prevention" OR	"Relapse"

	Terms])) OR (Relapses[MeSH Terms])) OR (Recrudescence[MeSH Terms])) OR (Recrudescences[MeSH Terms])		DE "Relapse (Disorders)"	
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4. Build a Search String

Use Boolean operators (AND, OR) and build a search string.

5. Search in PubMed, Scopus, Web of Science, Embase, and PsycINFO

Search string for:

PubMed:

```
((((((((Recurrences[MeSH Terms]) OR (Relapse[MeSH Terms])) OR
(Relapses[MeSH Terms])) OR (Recrudescence[MeSH Terms])) OR
(Recrudescences[MeSH Terms])) OR (((Recurrences[Title/Abstract]) OR
(Relapse[Title/Abstract])) OR (Relapses[Title/Abstract])))) AND
(((Encephalon[MeSH Terms]) OR (Brain[MeSH Terms])) OR
((Encephalon[Title/Abstract]) OR (Brain[Title/Abstract])))) AND
((((Epigenomic[MeSH Terms]) OR (Epigenetics[MeSH Terms])) OR
(Epigenetic[MeSH Terms])) OR (((Epigenomic[Title/Abstract]) OR
(Epigenetics[Title/Abstract])) OR (Epigenetic[Title/Abstract])))) AND
((((((((Drug Use Disorders[MeSH Terms]) OR (Substance Use Disorders[MeSH
Terms])) OR (Organic Mental Disorders, Substance-Induced[MeSH Terms])) OR
(Cheical Dependence[MeSH Terms])) OR (Disorder, Substance Related[MeSH
Terms])) OR (Abuse, Drug[MeSH Terms])) OR (Prescription Drug Abuse[MeSH
Terms])) OR (Drug Addiction[MeSH Terms])) OR (((Drug Use
Disorders[Title/Abstract]) OR (Substance Use Disorders[Title/Abstract])) OR
(Organic Mental Disorders, Substance-Induced[Title/Abstract])) OR (Prescription
Drug Abuse[Title/Abstract]))))
```

Scopus:

```
(TITLE-ABS-KEY("addict" OR "addiction, drug" OR "dependence, drug" OR
"drug addict" OR "drug addiction" OR "drug dependence model" OR "drug
dependency" OR "drug facilitation" OR "drug habituation" OR "drug physical
dependence" OR "drug usage disorder" OR "drug use disorder" OR "physical
```

dependence" OR "substance addiction" OR "substance dependence" OR "substance dependency" OR "substance usage disorder" OR "substance use disorder" OR "substance use disorders" OR "substance-related disorder" OR "substance-related disorders" OR "toxicomania" OR "toxicomanias" OR "toxicomanie" OR "drug dependence") AND TITLE-ABS-KEY("cerebra (brain)" OR "cerebrum" OR "encephalon" OR "brain") AND TITLE-ABS-KEY("Relapse") AND TITLE-ABS-KEY("Epigenetics"))

Web of Science:

((((TS=("addict" OR "addiction, drug" OR "dependence, drug" OR "drug addict" OR "drug addiction" OR "drug dependence model" OR "drug dependency" OR "drug facilitation" OR "drug habituation" OR "drug physical dependence" OR "drug usage disorder" OR "drug use disorder" OR "physical dependence" OR "substance addiction" OR "substance dependence" OR "substance dependency" OR "substance usage disorder" OR "substance use disorder" OR "substance use disorders" OR "substance-related disorder" OR "substance-related disorders" OR "toxicomania" OR "toxicomanias" OR "toxicomanie" OR "drug dependence")) AND TS=("Epigenetics")) AND TS=("cerebra (brain)" OR "cerebrum" OR "encephalon" OR "brain")) AND TS=("Relapse"))

Embase:

('addict'/exp OR 'addict' OR 'addiction, drug'/exp OR 'addiction, drug' OR 'dependence, drug'/exp OR 'dependence, drug' OR 'drug addict'/exp OR 'drug addict' OR 'drug addiction'/exp OR 'drug addiction' OR 'drug dependence model'/exp OR 'drug dependence model' OR 'drug dependency'/exp OR 'drug dependency' OR 'drug facilitation'/exp OR 'drug facilitation' OR 'drug habituation'/exp OR 'drug habituation' OR 'drug physical dependence'/exp OR 'drug physical dependence' OR 'drug usage disorder'/exp OR 'drug usage disorder' OR 'drug use disorder'/exp OR 'drug use disorder' OR 'physical dependence'/exp OR 'physical dependence' OR 'substance addiction'/exp OR 'substance addiction' OR 'substance dependence'/exp OR 'substance dependence' OR 'substance dependency'/exp OR 'substance dependency' OR 'substance usage disorder'/exp OR 'substance usage disorder' OR 'substance use disorder'/exp OR 'substance use disorder' OR 'substance use disorders'/exp OR 'substance use disorders' OR 'substance-related disorder'/exp OR 'substance-related disorder' OR 'substance-related disorders'/exp OR 'substance-related disorders' OR 'toxicomania'/exp OR 'toxicomania' OR 'toxicomanias'/exp OR 'toxicomanias' OR 'toxicomanie'/exp OR

'toxicomanie' OR 'drug dependence'/exp OR 'drug dependence') AND ('cerebra (brain)'/exp OR 'cerebra (brain)' OR 'cerebrum'/exp OR 'cerebrum' OR 'encephalon'/exp OR 'encephalon' OR 'brain'/exp OR 'brain') AND 'relapse'/exp AND 'epigenetics'/exp

PsycINFO:

XB (“addict” OR “addiction, drug” OR “dependence, drug” OR “drug addict” OR “drug addiction” OR “drug dependence model” OR “drug dependency” OR “drug facilitation” OR “drug habituation” OR “drug physical dependence” OR “drug usage disorder” OR “drug use disorder” OR “physical dependence” OR “substance addiction” OR “substance dependence” OR “substance dependency” OR “substance usage disorder” OR “substance use disorder” OR “substance use disorders” OR “substance-related disorder” OR “substance-related disorders” OR “toxicomania” OR “toxicomanias” OR “toxicomanie” OR “drug dependence”) AND XB (“Epigenetics”) AND XB (“cerebra (brain)” OR “cerebrum” OR “encephalon” OR “brain”) AND XB (“cerebra (brain)” OR “cerebrum” OR “encephalon” OR “brain”)

6. Record and export the results.

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2026

PUBLICATION DATE

1 year

5 years

10 years

Custom Range

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63 results

Page 1 of 7

Cell type-specific **epigenetic** priming of gene expression in nucleus accumbens by cocaine.

Cite

Mews P, Van der Zee Y, Gurung A, Estill M, Futamura R, Kronman H, Ramakrishnan A, Ryan M, Reyes AA, Garcia BA, Browne CJ, Sidoli S, Shen L, Nestler EJ. Sci Adv. 2024 Oct 4;10(40):eado3514. doi: 10.1126/sciadv.ado3514. Epub 2024 Oct 4. PMID: 39365860 **Free PMC article.**

A hallmark of addiction is the ability of drugs of abuse to trigger **relapse** after periods of prolonged abstinence. Here, we describe an **epigenetic** mechanism whereby chronic cocaine exposure causes lasting chromatin and downstream transcriptional modifications in the ...

Substance use disorder a bio-directional subset of reward deficiency syndrome.

Cite

Blum K, Gold M, Demetrovics Z, Archer T, Thanos PK, Baron D, Badgaiyan RD. Front Biosci (Landmark Ed). 2017 Jun 1;22(9):1534-1548. doi: 10.2741/4557. PMID: 28410131 **Free article.**

Like a "double edge sword" SUD has a biological bi-directional (bio-directional) effect on the **brain** reward circuitry. The acute intake of psychoactive drugs results in heightened dopaminergic activity, while, the opposite, hypodopaminergia occurs following chronic abuse. ...

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Neuroscience29

Pharmacology, Toxicology and Pharmaceutics18

Biochemistry, Genetics17

Document title

Authors

Source

Year

Citations

1

Histone Deacetylase 5 in Prelimbic Prefrontal Cortex Limits Context-Associated Cocaine Seeking

Barry S.M., Huebschman J.L., Devries D.M., Penrod R.D., Cowan C.W.

Biological Psychiatry, 99(6)

2026

2

2

Preclinical evidence for statin modulation of analgesic tolerance and substance use disorders: a narrative review

Ranaei M.S., Seyedhosseini Tamijani S.M., Ghazvini H., Alaedini Shourmasti K., Rafeiee R.

American Journal of Drug and Alcohol Abuse, 52(2)

2026

1

3

The imprisoned brain: An integrated neurobiological view of substance dependence

Bacchi A.D.

Advanced Neurology, 5(3), 025090018

2026

1

Web of Science:

webofscience.com/wos/woscc/summary/Sbafc99e-029e-4017-9053-73dbc7ec458d-01c7da634b/date-descending/1

Keywords: < + substance use disorder + epigenetics + addiction >

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1 Precision medicine in substance use disorders: Integrating behavioral, environmental, and biological insights

Guerrin, CG.; Tesselaar, DRM; (...); Homberg, JR

Sep 2025 | NEUROSCIENCE AND BIOBEHAVIORAL REVIEWS 176

Substance use disorders (SUD) are chronic, relapsing conditions marked by high variability in treatment response and frequent relapse. This variability arises from complex interactions among behavioral, environmental, and biological factors unique to each individual. Precision medicine, which tailors treatment to patient-specific char ... Show more

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10 Citations 369 References

2 The Role of Oxytocin and Oxytocin Gene Receptor Methylation During Withdrawal Therapy in Males With Alcohol Use Disorder

Proskynitopoulos, PJ; Haarmeyer, AF; (...); Rhein, M

Jul 14 2025 | ADDICTION BIOLOGY 39(7)

Enriched Cited References

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Review • Embase, MEDLINE, NURSING

1. Preclinical evidence for statin modulation of analgesic tolerance and substance use disorders: a narrative review

Ranaiv M.S., Seyedhosseini Tamijani S.M., Ghazvini H., Alaedini Shourmasti K., Rafeaie R.

American Journal of Drug and Alcohol Abuse 2026 52:2 (183-198)

Abstract Index terms Full text Similar records

Article • Embase

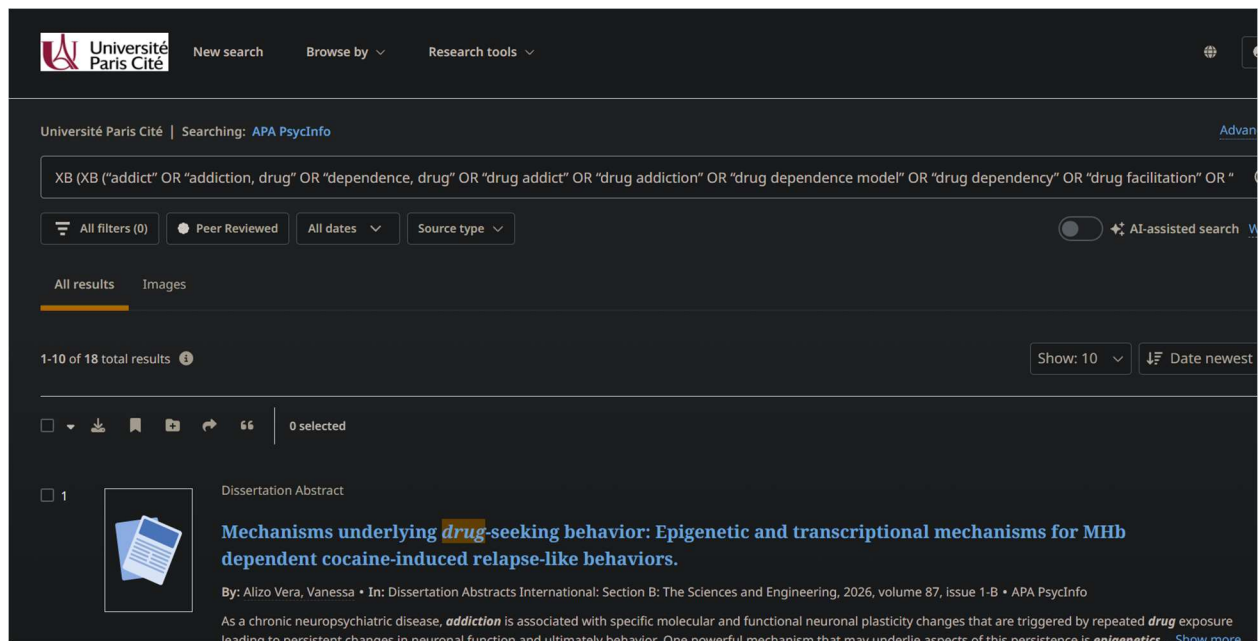
2. Decoding Addiction: Bridging Neuroscience, Mental Health, and Therapeutics to Redefine Substance Use Narratives

Siddiqi K.N., Dar M.A., Kanyu S.M., Naik S.A.

International Journal of Current Pharmaceutical Review and Research 2025 17:4 (1340-1346)

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PsycINFO:

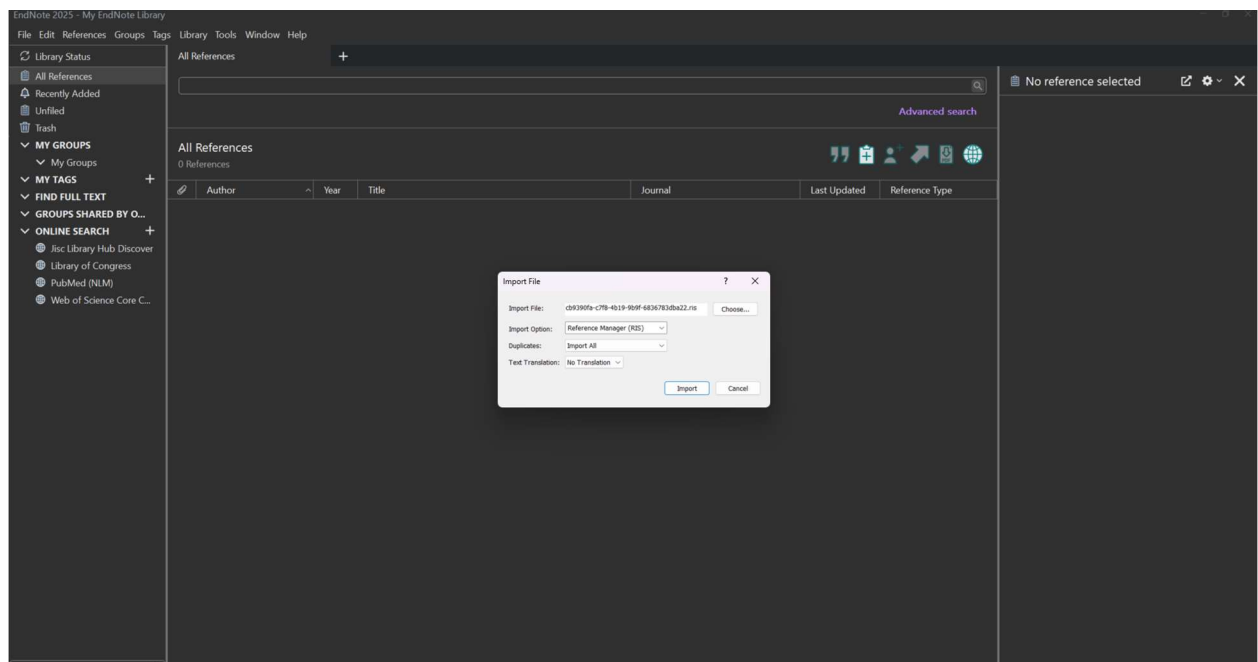


Part 2

1. Import References into EndNote

Take a screenshot of each step during the process.

1-



2-

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Library Status

File	Journal	Last Updated	Reference Type
Epigenetics and drug addiction: a focus on MeCP2 and ...	Med Sci (Paris)	20/06/1405	Journal Article
ipigenetics and drug addiction: a focus on MeCP2 and ...	M S-MEDICINE SCIENCES	20/06/1405	Journal Article
ipigenetics and drug addiction: A focus on MeCP2 and ...	Medicine/Sciences	20/06/1405	Journal Article
Multi-omics profiling of DNA methylation and gene exp...	Transl Psychiatry	20/06/1405	Journal Article
role of prefrontal cortex in the extinction of drug memo...	Psychopharmacology (Berl)	20/06/1405	Journal Article
role of prefrontal cortex in the extinction of drug memo...	Psychopharmacology	20/06/1405	Journal Article
role of prefrontal cortex in the extinction of drug memo...	Psychopharmacology	20/06/1405	Journal Article
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Neuronal extracellular signal-regulated kinase (ERK) activ...	Frontiers in Integrative Neuroscience	20/06/1405	Journal Article
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EPiGENETIC CONTROL OF AN AUXILIARY SUBUNIT OF V...	Neuropsychopharmacology	20/06/1405	Journal Article
Limitations of the protective measure theory in explaini...	Eating and Weight Disorders	20/06/1405	Journal Article
PET imaging for addiction medicine: From neural mecha...	Progress in Brain Research	20/06/1405	Book Section
Ubiquitin-proteasomal regulation of chromatin remodel...	Sci Adv	20/06/1405	Journal Article
Werner, Craig T.; Gancarz, A.M.; Dietz, D.M. 2019	Mechanisms regulating compulsive drug behaviors	20/06/1405	Book Section
Wang, Z. J.; Martin, J. A.; ... 2016	BRG1 in the Nucleus Accumbens Regulates Cocaine-See...	20/06/1405	Journal Article
Wang, Q.; Qin, F.; Wang... 2023	Normalization of the H3K9me2/H3K14ac-ΔfosB pathwa...	20/06/1405	Journal Article
Wang, Q.; Qin, F.; Wang... 2023	Normalization of the H3K9me2/H3K14ac-ΔfosB pathwa...	20/06/1405	Journal Article
Wang, B.; Wang, J. L.; Be... 2025	Cell-type specific epigenetic and transcriptional mech...	20/06/1405	Journal Article
Wang, B.; Wang, J. L.; Be... 2025	Cell-type specific epigenetic and transcriptional mech...	20/06/1405	Journal Article
Walsh, Anthony; Johnso... 2012	Drugs, crime, and the epigenetics of hedonic allostasi...	20/06/1405	Journal Article
Vassoler, F. M.; Oliver, D... 2017	Transgenerational attenuation of opioid self-administr...	20/06/1405	Journal Article

2019 #... Summary Edit PDF

Mechanisms regulating compulsive drug behaviors

Werner, C.T., Gancarz, A.M. & Dietz, D.M.

Neural mechanisms of addiction.

2019

Pages 137-155

DOI: 10.1016/B978-0-12-812202-0.00010-5

Links

<https://research.elsevier.com/jlink/48433484-82df-3da0-ae19-9f7037d33908>

Abstract

Compulsive drug seeking and use are hallmark symptoms of substance use disorders. These behaviors emerge as the brain undergoes changes in circuits and signaling pathways that lead to dysfunctional motivation and reward processing, which ultimately contribute to maintaining persistent relapse vulnerability. Preclinical models of compulsion...

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 - Library of Congress
 - PubMed (NLM)
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All References

164 References

Author	Year	Title	Journal	Last Updated	Reference Type
Zhang, W. H.; Cao, K. X... 2019	2019	Role of prefrontal cortex in the e	Frontiers in Cellular Neuroscience	20/06/1405	Journal Article
Zamora-Martinez, E. R.; ... 2014	2014	Neuronal extracellular signal-reg	Frontiers in Integrative Neuroscience	20/06/1405	Journal Article
You, C.; Vandegriff, B. J... 2018	2018	Histone Deacetylase Inhibitor Su	Alcohol Clin Exp Res	20/06/1405	Journal Article
Yang, J.; Dai, Y.; Li, J... 2026	2026	Research Progress on the Regula	Addict Biol	20/06/1405	Journal Article
Wu, T.; Xu, W.; Lin, Z.; X... 2025	2025	Gamma-aminobutyric acid A rec	Neuroreport	20/06/1405	Journal Article
Wright, K. N.; Dossat, A... 2018	2018	Optogenetic inhibition of medial	Integr Zool	20/06/1405	Journal Article
Wood, D. J.; Tsvetkov, E... 2025	2025	Epigenetic Control of an Auxilia	Biol Psychiatry	20/06/1405	Journal Article
Wood, D. J.; Tsvetkov, E... 2024	2024	EPiGENETIC CONTROL OF AN AI	Neuropsychopharmacology	20/06/1405	Journal Article
Wiss, David A.; Brewet... 2022	2022	Limitations of the protective me	Eating and Weight Disorders	20/06/1405	Journal Article
Wiers, C. E.; Cabrera, E... 2016	2016	PET imaging for addiction med	Progress in Brain Research	20/06/1405	Book Section
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Wang, B.; Wang, J. L.; Be... 2025	2025	Cell-type specific epigenetic and transcriptional mech...	Frontiers in Cellular Neuroscience	20/06/1405	Journal Article
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Vassoler, F. M.; Oliver, D... 2017	2017	Transgenerational attenuation of opioid self-administr...	Neuropharmacology	20/06/1405	Journal Article
Van De Wiel, S. M. W.; V... 2014	2014	Designing modulators of 5-hydroxytryptamine signaling...	Expert Opin Drug Discovery	20/06/1405	Journal Article
van de Wiel, S. M.; Verh... 2014	2014	Designing modulators of 5-hydroxytryptamine signaling...	Expert Opin Drug Discov	20/06/1405	Journal Article
Thanekar, R.; Iakande... 2026	2026	Oxytocin as a neurotherapeutic target for nicotine withdr...	Neuropeptides	20/06/1405	Journal Article
Ianiguchi, M.; Carneir... 2016	2016	HDACs and NPAS4 regulate cocaine reward environment...	Neuropsychopharmacology	20/06/1405	Journal Article
Stewart, A. F.; Lepack, A... 2023	2023	Histone H3 dopaminylation in nucleus accumbens, but n...	Mol Cell Neurosci	20/06/1405	Journal Article
Starkman, B. G.; Sakhar... 2012	2012	Epigenetics-beyond the genome in alcoholism	Alcohol Res	20/06/1405	Journal Article
Spanagel, R.; Durstewitz... 2013	2013	A systems medicine research approach for studying alco...	Addict Biol	20/06/1405	Journal Article
Spanagel, R. 2015	2015	A systems medicine approach towards understanding al...	Alcohol and Alcoholism	20/06/1405	Journal Article

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2025 #... Summary Edit PDF

Cell-type specific epigenetic and transcriptional mechanisms in substance use disorder

Wang, B., Wang, J.L., Beacher, N.J., Lin, D.T. & Zhang, Y.

FRONTIERS IN CELLULAR NEUROSCIENCE

2025

DOI: 10.3389/fncel.2025.1552032

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Abstract

Substance use disorder (SUD) is a chronic and relapse-prone neuropsychiatric disease characterized by impaired brain circuitry within multiple cell types and neural circuits. Recent advancements in single-cell transcriptomics, epigenetics, and neural circuit research have unveiled molecular and cellular alterations associated with SUD. These studies have provided...

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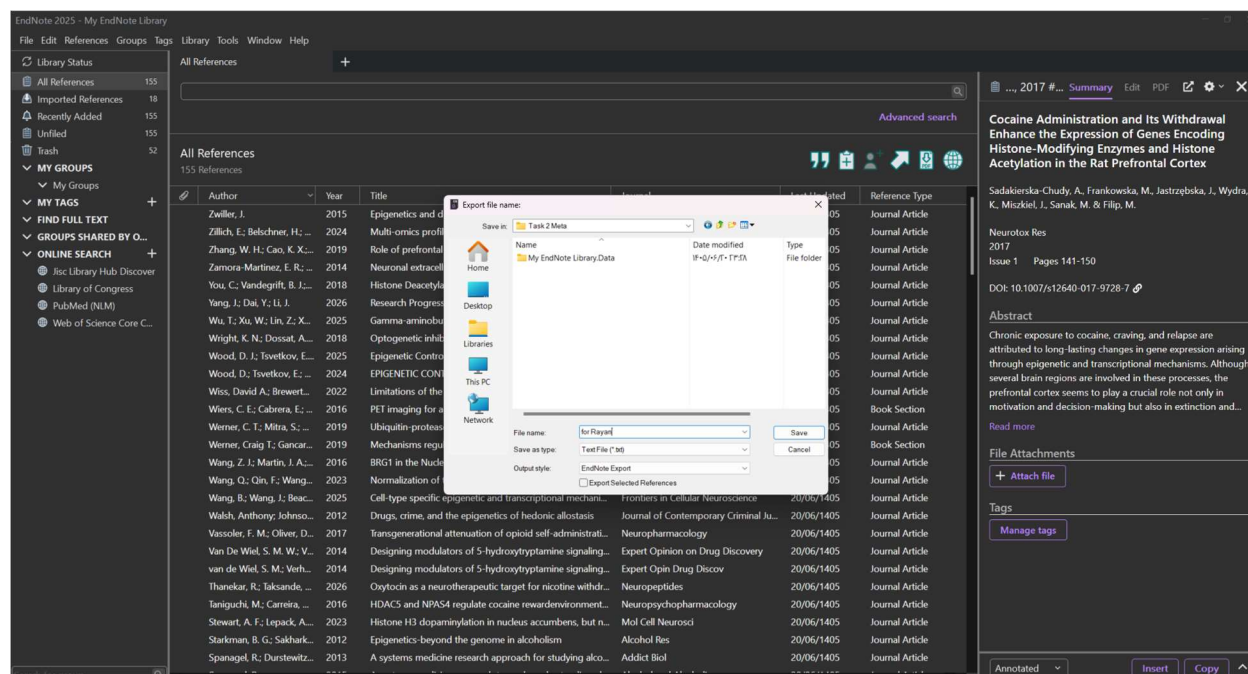
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4-



2. Remove Duplicates

Find duplicates both automatically and manually. Take one screenshot for each.

Record:

- Number of duplicates removed automatically **5 (above)**
- Number of duplicates removed manually **4 (above)**

3. Find Missing Abstracts

Find missing abstracts both automatically and manually. Take one screenshot for each.

The screenshot shows the EndNote interface with a list of 155 references. The 'Abstract' column for several entries is empty, indicating missing abstracts. The right-hand pane shows the abstract for the selected entry, 'Optogenetic inhibition of medial prefrontal cortex projections to the nucleus accumbens (NAc) are critically important for reinstated cocaine-seeking in a rodent model of cocaine relapse'.

Author	Year	Title	Journal	Last Updated	Reference Type	Abstract
Zillich, E.; Belschner, H.; ...	2024	Multi-omics profiling of DNA methylation and gene exp...	Transl Psychiatry	20/06/1405	Journal Article	DNA methylation cocaine-primed reinstatement corticolimbic methionine optogenetics
Zhang, W. H.; Cao, K. X.; ...	2019	Role of prefrontal cortex in the extinction of drug memo...	Psychopharmacology	20/06/1405	Journal Article	DNA methylation has been identified as a powerful and activity-dependent regulator of changes in the brain that may underlie neuroadaptations in response to various types of stimuli, including exposure to drugs of abuse. Indeed, the medial prefrontal cortex (mPFC) projections to the nucleus accumbens (NAc) are critically important for reinstated cocaine-seeking in a rodent model of cocaine relapse. This circuitry undergoes several epigenetic modifications following cocaine exposure, including changes in DNA methylation that are associated with drug-seeking behavior. We have previously shown that methyl supplementation via L-Methionine (MET) administration attenuates cocaine-seeking behavior and reverses expression and methylation patterns of the immediate early gene c-fos, suggesting that MET may act by altering the excitability of this circuitry during cocaine reinstatement. In the current study, male rats were microinjected with an adeno-associated virus overexpressing halorhodopsin in the mPFC, optical fibers were surgically implanted into the NAc, and the rats were given injections of MET daily. Rats underwent acquisition of cocaine self-administration (0.75

The screenshot shows the EndNote interface with a search for missing abstracts. A dialog box indicates 'No matching references found.' The right-hand pane shows the abstract for the selected entry, 'Optogenetic inhibition of medial prefrontal cortex projections to the nucleus accumbens core and methyl supplementation via L-Methionine attenuates cocaine-primed reinstatement'.

Author	Year	Title	Journal	Last Updated	Reference Type	Abstract
Zwiller, J.	2015	Epigenetics and drug addiction: A focus on MeCP2 and ...	Medicine/Sciences	20/06/1405	Journal Article	Optogenetic inhibition of medial prefrontal cortex projections to the nucleus accumbens core and methyl supplementation via L-Methionine attenuates cocaine-primed reinstatement
Zillich, E.; Belschner, H.; ...	2024	Multi-omics profiling of DNA methylation and gene exp...	Transl Psychiatry	20/06/1405	Journal Article	DNA methylation cocaine-primed reinstatement corticolimbic methionine optogenetics

Record:

- Number of references with missing abstracts 0
- Number completed automatically 0
- Number completed manually 0

4. Export the Final Library

Send the exported .txt file.

Part3

1. Create a Rayyan Account

- Go to <https://www.rayyan.ai>.
- Sign up using your email (or Google/Gmail account).
- After logging in, complete your full name.

2. Add Your Instructor (Me) as a Collaborator on the Project

- After creating your project (named Systematic_Screening_[Your_full name]), go to "Invite Collaborators".
- Enter the instructor's email address: B*****ahar*****@*****om
- Assign "Editor" or "Reviewer" access so I can monitor your progress and provide feedback.

3. Import All Articles from Your EndNote Library into Rayyan

The screenshot displays the Rayyan AI web interface for a project titled "named Systematic_Screening_[Hamid Reza Ghahramani]". The interface includes a top navigation bar with tabs for Overview, Review data, Screening, Full text screening, Data extraction, and Risk of bias. Below the navigation bar, there's a section for "Review Info" with details like Review Title, Review Type (Meta-analysis), Review Domain (Other), and a Description. The "Data Summary" section shows the following statistics:

Imported References	Total Duplicates	Unresolved	Resolved	Not Duplicate	Deleted
155	0	0	0	0	0

Buttons for "Add References", "Detect Duplicates", "Start Resolving", and "PRISMA" are visible. The "Review Members" section is partially visible at the bottom.

4. Screen 100 of Imported Articles

- Review **articles** based on **title and abstract** as follows:
 - **Included:** Articles that appear relevant to your research question.
 - **Excluded:** Articles that are clearly irrelevant.
 - **Maybe:** For cases that need further examination.
-

5. Requirement: At Least 5 Included Articles in the First Stage

- You must mark at least 5 articles as **Included** at this stage.
-

6. Obtain and Attach Full-Text PDFs for All Included Articles

- For **all articles** marked as Included in the previous step (at least 5, but preferably all Included articles), obtain the **PDF/full-text** via your university, or library request.
-

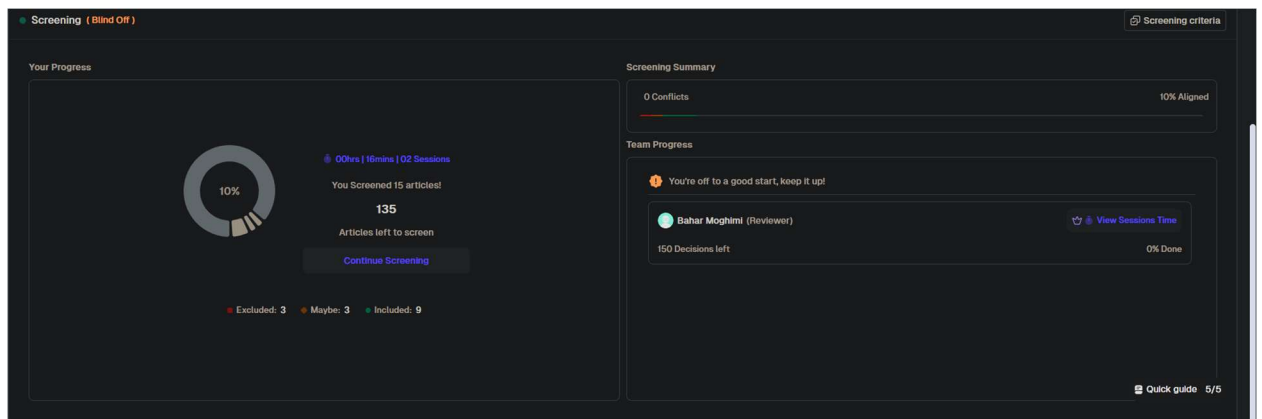
7. Make a Final Decision Based on Full-Text Reading

- Read **each** Included article completely and make a final decision based on your pre-defined inclusion/exclusion criteria:
 - **Confirm Included** (final acceptance for the review)
 - **Reject (Exclude)** (final exclusion due to mismatch with criteria)
-

8. Documentation and Final Report (Rayyan Section)

Your report must include:

- **Screenshot** of your Rayyan project showing **total number of articles**, number of Included, Excluded, and Maybe.



- **Screenshot of the list of Included articles**

This screenshot shows the article list and details for 'Epigenetics and drug addiction: A focus on MeCP2 and on histone acetylation'. The left sidebar lists 9 articles with checkboxes and status icons (green check for included, red X for excluded, yellow question mark for maybe). The main panel shows the article title, ID (687834372), and a PDF icon. Below the title, it says 'Author and journal details are hidden for this review' with a 'Show' link. A 'See more' link is also present. The 'Abstract' section is expanded, showing the full text. At the bottom, there are buttons for 'Included', 'Maybe', 'Exclude', 'Reason', 'Label', and 'Analyze'. The right sidebar contains 'Filters' for 'Keywords for include' and 'Keywords for exclude', each with a list of terms and counts.

Article List (Left Sidebar):

ID	Date	Title	Status
01	2015-01-01	Epigenetics and drug addiction: A focus on MeCP2 and on histone acetylation	Included
02	2024-10-09	Multi-omics profiling of DNA methylation and gene expression a...	Included
03	2019-01-01	Role of prefrontal cortex in the extinction of drug memories	Maybe
04	2014-01-01	Neuronal extracellular signal-regulated kinase (ERK) activity as ...	Excluded
05	2018-11-01	Histone Deacetylase Inhibitor Suberanilohydroxamic Acid Treat...	Excluded
06	2025-09-01	Research Progress on the Regulatory Mechanisms of Gut Micro...	Included
07	2025-12-10	Gamma-aminobutyric acid A receptor alpha 4 in the nucleus acc...	Included
08	2018-11-01	Optogenetic inhibition of medial prefrontal cortex projections to ...	Included
09	2025-10-15	Epigenetic Control of an Auxiliary Subunit of Voltage-Gated Sod...	Included

Article Details (Main Panel):

Title: Epigenetics and drug addiction: A focus on MeCP2 and on histone acetylation

ID: 687834372

Abstract: Chronic drug exposure alters gene expression in the brain, which is believed to underlie compulsive drug seeking and drug taking behavior. Recent evidence shows that drug-induced long-term neuroadaptations in the brain are mediated in part by epigenetic mechanisms. By remodeling chromatin, this type of regulation contributes to drug-induced synaptic plasticity that translates into behavioral modifications. How drug-induced alterations in DNA methylation regulate gene expression is reviewed here, with a focus on MeCP2, a protein binding methylated DNA. The importance of histone modifications, especially acetylation is also discussed, with an emphasis on the effects of inhibitors of histone deacetylases on drug-induced behavioral changes. The precise identification of the epigenetic mechanisms that are under the control of drugs of abuse may help to uncover novel targets for the treatment of drug seeking and relapse. © 2015 médecine/sciences - Inserm.

Buttons: Included, Maybe, Exclude, Reason, Label, Analyze

Filters (Right Sidebar):

Keywords for include	Keywords for exclude
Select All	Select All
compared with	rats
CCT	this review
RCT	animal
trial	rat
placebo	animals
randomly	cells
crossover	mice
cross over	mouse
randomised	rodent
randomized	in vitro